MPG No. 19.1.68 $3m$ (3m1 setting) [Type I, trigonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{xy} = \mathbb{T}_z^{(1)}$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ -M_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -M_{xxx} & 0 \end{bmatrix}$$

$M_{xxx} = \mathbb{M}_{3c}^{(1)}$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$M_{yzx} = 3\mathbb{T}_u^{(1)}$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$
$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxzx} \\ M_{xxzx} & -M_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$M_{xxzx} = \mathbb{M}_{4c}^{(1)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yzzx} & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & -M_{yzzx} & 0 & 0 \\ -M_{xxxy} & M_{xxxy} & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = -2\mathbb{T}_{3s}^{(1)}$$
$$M_{xxxy} = 2\mathbb{T}_{z}^{(1)} - 2\mathbb{T}_{z}^{\alpha(1)}$$
$$M_{yzzx} = -\mathbb{T}_{z}^{(1)} - 4\mathbb{T}_{z}^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_{z}^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & M_{xxxy} \\ 0 & -M_{xxzx} & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{T}_{3s}^{(2)}$$
$$M_{xxxy} = \mathbb{T}_{z}^{(3)} - \mathbb{T}_{z}^{\alpha(2)}$$
$$M_{zzxy} = \mathbb{T}_{z}^{(3)} + 2\mathbb{T}_{z}^{(4)} + 2\mathbb{T}_{z}^{\alpha(2)}$$
$$M_{yzzx} = \mathbb{T}_{z}^{(4)} - \mathbb{T}_{z}^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = \mathbb{T}_{z}^{(6)} - \mathbb{T}_{z}^{\alpha(3)}$$
$$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$$
$$M_{xyxx} = \mathbb{T}_{z}^{(5)} - \mathbb{T}_{z}^{\alpha(3)}$$
$$M_{xyzz} = \mathbb{T}_{z}^{(5)} + 2\mathbb{T}_{z}^{(6)} + 2\mathbb{T}_{z}^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} & 0 & M_{xxxz} & M_{xxyx} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxyx} & 0 & -M_{xxxz} & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} & 0 & M_{yzxz} & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & -M_{yzxz} & 0 & 0 & -M_{yzzx} & 0 & 0 \\ -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} & \frac{M_{xxxy}}{2} + \frac{M_{xxyx}}{2} & 0 & -M_{xxxz} & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$M_{xxzx} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} + \mathbb{T}_{3s}^{(2)}$$
$$M_{xxxy} = 2\mathbb{T}_{z}^{(1)} + \mathbb{T}_{z}^{(3)} - 2\mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\alpha(2)}$$
$$M_{xxxz} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} - \mathbb{T}_{3s}^{(2)}$$
$$M_{xxyx} = 2\mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{(3)} - 2\mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\alpha(2)}$$
$$M_{zzxy} = \mathbb{T}_{z}^{(3)} + 2\mathbb{T}_{z}^{(4)} + 2\mathbb{T}_{z}^{\alpha(2)}$$
$$M_{yzzx} = -\mathbb{T}_{z}^{(1)} + \mathbb{T}_{z}^{(4)} - 4\mathbb{T}_{z}^{\alpha(1)} - \mathbb{T}_{z}^{\alpha(2)}$$
$$M_{yzxy} = -\mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)}$$
$$M_{yzxz} = -\mathbb{T}_{z}^{(1)} - \mathbb{T}_{z}^{(4)} - 4\mathbb{T}_{z}^{\alpha(1)} + \mathbb{T}_{z}^{\alpha(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzzx} & M_{yzxy} & 0 & M_{yzxz} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzxz} & 0 & 0 & M_{yzzx} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$$
$$M_{yzxz} = \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$
$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -M_{xxxz} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxxz} & 0 & 0 \\ 0 & 0 & -M_{xxxz} \\ 0 & -M_{xxxz} & 0 \end{bmatrix}$$

$$M_{xxxz} = \mathbb{M}_{4c}^{(1)}$$